SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:		**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>*.
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Title: Applying Advanced Agile Methodologies and Project Management Techniques to Manage Software Projects Effectively

1. Project Vision and Stakeholder Identification

1.1 Project Vision

- Nowadays, many people find it difficult to manage their income and daily expenses because they do not maintain proper financial records. Spending on food, travel, shopping, and bills without planning can lead to overspending. The Personal Expense and Budget Management System aims to solve this problem by providing an application to record income and expenses, categorize spending, create monthly budgets, receive alerts, and view financial reports. It helps users understand their spending habits and manage their money more effectively.

1.2 Project Objectives

- The main objective of this project is to help users maintain accurate financial records and control their spending. The system allows users to record daily transactions, organize expenses into categories, set monthly budgets, receive alerts when spending approaches the budget limit, and generate reports showing total income, expenses, and savings. These features help users plan their monthly expenses and make better financial decisions.

1.3 Stakeholder Identification

- The main stakeholders are end users, the product owner, the development team, the Scrum Master, and the software tester. End users record transactions and manage budgets, while the product owner identifies and prioritizes requirements. The development team designs and builds the application, the Scrum Master coordinates Agile activities, and the tester checks the functionality and accuracy of calculations.

2. Create Epics and User Stories

- Epics represent the major functionalities of the application, while user stories describe specific features from the user's point of view. The main epics include account management, transaction management, expense categorization, budget planning, budget alerts, and financial reporting. These epics divide the project into smaller requirements that can be developed and tested during different sprints.

Epic	**User story**
Account Management	As a user, I want to log in so that my financial records remain private.
Transaction Management	As a user, I want to record income and expenses so that I can track my money.
Expense Categorization	As a user, I want to categorize expenses so that I can understand my spending.
Budget Planning	As a user, I want to set monthly budgets so that I can control expenses.
Budget Alerts	As a user, I want to receive spending alerts so that I can avoid overspending.
Financial Reports	As a user, I want to view monthly reports so that I can analyze my spending habits.

3. Define Acceptance Criteria

- Acceptance criteria describe the conditions that a feature must satisfy to be considered complete. For the login feature, valid credentials should allow the user to access the dashboard, while incorrect credentials should display an error message. For transaction management, valid income and expense details must be saved, and the totals must update correctly. When a user sets a budget, the system should display the spending limit and remaining amount. If spending reaches 80% of the budget, an alert should be displayed. Financial reports should show the selected month's income, expenses, balance, and category-wise spending.

4. Prioritize the Product Backlog

- The product backlog contains all the features planned for development, arranged according to their importance and dependency. User login is given the highest priority because users need secure access to their records. Recording income and expenses comes next, followed by balance calculation and budget creation. Budget alerts and financial reports are prioritized after the basic transaction features because they depend on accurate financial data. This order helps the team develop the essential features first and add advanced functionality later.

Priority	Feature	Reason
1	User login	Provides account access.
2	Record income	Tracks money received.
3	Record expenses	Tracks daily spending.
4	View balance	Shows the financial position.
5	Set budgets	Helps control spending.
6	Budget alerts	Warns about overspending.
7	Financial reports	Helps analyze spending patterns.

5. Plan the Sprint Backlog

- The project follows a two-week sprint cycle. During Sprint 1, the team focuses on developing the basic application features, including login, income recording, expense recording, and balance calculation. The development tasks include designing forms, creating the database, validating user inputs, storing transactions, calculating totals, and testing the results.

User story	Main development tasks	Story points
Login	Design form and validate credentials.	5
Record income	Create form and save income details.	3
Record expenses	Create expense form and categories.	5
View balance	Calculate income minus expenses.	3
Total		16

6. Estimate Effort Using Story Points

- Story points are used to estimate the relative effort and complexity of each user story. The team uses the Fibonacci sequence of 1, 2, 3, 5, 8, and 13 to assign estimates. Login is estimated at 5 points because it involves authentication and validation. Income recording receives 3 points, while expense recording receives 5 points because it includes category selection and transaction storage. Budget creation and alerts receive 5 points each, financial reports receive 8 points, and balance calculation receives 3 points.
- For the first sprint, the selected stories have estimates of 5, 3, 5, and 3 points. Therefore, the total planned effort is:

$$[5+3+5+3=boxed{16\text{ story points}}]$$

- This estimate helps the team plan its workload, although actual completion depends on team capacity and development challenges.

7. Sprint Goal and Expected Deliverables

- The goal of Sprint 1 is to develop a working application that allows users to log in, record income and expenses, and view their remaining balance. By the end of the sprint, the team expects to deliver a login module, income and expense entry forms, a transaction database, and a dashboard displaying financial totals. The team will also test transaction validation and verify that balance calculations are correct before demonstrating the application.
- The remaining features will be developed in later sprints. Sprint 2 will focus on editing transactions and creating budgets, Sprint 3 will introduce budget alerts and financial reports, and Sprint 4 will focus on testing, improvements, and final integration. A feature will be considered complete when it has been implemented, tested, integrated, and verified to work correctly.

Conclusion

- The Personal Expense and Budget Management System provides a practical way for users to track income, manage expenses, and control monthly spending. Agile sprint planning divides the application into manageable tasks, allowing the development team to build and test essential features first. The iterative approach also makes it easier to improve the application based on feedback and deliver a useful system.